

Assignment - 2 Output

AI-Powered Customer Support Automation System

Verification Screenshots & Output Evidence (LangGraph)

System Environment: Ollama Local LLM (qwen2.5:0.5b / llama3.1)

Memory Database: SQLite Persistent Checkpointer (memory.db)

RAG Indexing: TF-IDF Fallback Vector Store

1. Agent Routing

Evidence of the Intent Classification node routing each query to the correct department (Sales, Technical, Billing, or Account).

```

--- Running Demo Query 1 of 5 ---
=====
CUSTOMER: "What are the pricing plans available for your software?"
=====

[Node: Classify Intent] classified query: 'What are the pricing plans available for your software?' -> Category: Sales (Rule-based)
Ollama embedding failed for text: What are the pricing plans ava... Error: This server does not support embeddings. Start it with '--embeddings' (status code: 500)
Retrieval falling back to TF-IDF due to: This server does not support embeddings. Start it with '--embeddings' (status code: 500)
[Node: Sales Agent] Retrieved context from: faq_document.txt, pricing_guide.txt
[Node: Supervisor Evaluation] Polished and validated final response.
=====
FINAL SUPPORT RESPONSE:
The pricing plans available for ABC Technologies' software include:

1. Basic ($19/month) - This plan is ideal for small startups as it provides a basic level of support and features.
2. Professional ($49/month) - Ideal for growing businesses, this plan offers more advanced functionalities and features.
3. Enterprise (custom pricing starting at $199/month, billed annually) - Suitable for larger organizations or those who require custom solutions.
=====

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2. Human-in-the-Loop

Query 4 (refund request) triggers the supervisor review interruption, pausing graph execution. Once the supervisor inputs approval ('y'), execution resumes and generates the response.

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--- Running Demo Query 2 of 5 ---
=====
CUSTOMER: "I forgot my account password."
=====

[Node: Classify Intent] classified query: 'I forgot my account password.' -> Category: Account (Rule-based)
Ollama embedding failed for text: I forgot my account password.... Error: This server does not support embeddings. Start it with '--embeddings' (status code: 500)
Retrieval falling back to TF-IDF due to: This server does not support embeddings. Start it with '--embeddings' (status code: 500)
[Node: Account Agent] Retrieved context from: technical_manual.txt, faq_document.txt
[Node: Supervisor Evaluation] Polished and validated final response.
=====
FINAL SUPPORT RESPONSE:
To reset your password, follow these steps:

1. Go to the login page of ABC Technologies.
2. Click on "Forgot Password" link provided below the input fields.
3. Enter your registered email address and click "Send Link".
4. An email with a secure password reset link (valid for 1 hour) will be sent to you.

If the reset email is not received, please check your spam or promotions folder, add support@abctechnologies.com to your address book, and wait 5 minutes before trying again.
=====

```

3. RAG Retrieval

Demonstrates the system loading relevant context blocks from company text documents (faq_document.txt, pricing_guide.txt, technical_manual.txt, company_policy.txt) to ground answers in verified information.

```
--- Running Demo Query 3 of 5 ---
=====
CUSTOMER: "My application crashes whenever I upload a file."
=====

[Node: Classify Intent] classified query: 'My application crashes whenever I upload a file.' -> Category: Technical (Rule-based)
Ollama embedding failed for text: My application crashes whenever I upload a file... Error: This server does not support embeddings. Start it with '--embeddings' (status code: 500)
Retrieval falling back to TF-IDF due to: This server does not support embeddings. Start it with '--embeddings' (status code: 500)
[Node: Technical Agent] Retrieved context from: technical_manual.txt, faq_document.txt
[Node: Supervisor Evaluation] Polished and validated final response.
=====

FINAL SUPPORT RESPONSE:
Based on the information provided in the technical manual and FAQ document, here are some steps you can take to troubleshoot this issue:

1. Check File Size Limit: Ensure that your file size is below 25MB. If it exceeds this limit, the application will crash when trying to upload a file.

2. Verify Extension: Confirm that the extension of the uploaded file matches one of the supported formats (PDF, CSV, PNG, JPG). Incorrect extensions can cause the application to crash or fail to process the file correctly.

3. Clear Browser Cache and Try Again: Clearing your browser cache and trying again may resolve any issues with the file upload functionality.

4. Update Client Software: Ensure that you are using the latest version of our desktop client software. Older versions might have known bugs or compatibility issues, which could be causing the application to crash when uploading files.

5. Check for Corrupted Files: If the above steps do not resolve the issue, it may be necessary to check if there are any corrupted files in your system that could be causing the problem. You can try opening the file locally first before attempting to upload it again.

6. Contact Support: If none of these solutions work, you might want to contact our technical support team for further assistance.
=====
```

4. Memory Storage & Recall

Query 5 ('What was my previous support issue?') extracts past session history from SQLite and accurately summarizes that the previous issue was a file upload crash (from Query 3).

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--- Running Demo Query 4 of 5 ---
=====
CUSTOMER: "I need a refund for my annual subscription."
=====

[Node: Classify Intent] classified query: 'I need a refund for my annual subscription.' -> Category: Billing (Rule-based)
Ollama embedding failed for text: I need a refund for my annual ... Error: This server does not support embeddings. Start it with '--embeddings' (status code: 500)
Retrieval falling back to TF-IDF due to: This server does not support embeddings. Start it with '--embeddings' (status code: 500)
[Node: Billing Agent] Retrieved context from: company_policy.txt, faq_document.txt

!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
!!! DANGER / HIGH-RISK ACTION DETECTED !!!
Supervisor Approval Required for action in category: Billing
Agent's Draft Response:
Certainly! To initiate the refund process, please provide me with the details of your annual subscription, including the date of purchase or billing date. Once I have this information, I can assist you in processing the refund according to our refund policy. Please let me know if there are any specific steps or forms required for this process.
!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!

Do you approve this response? (y/n/exit): y
Resuming graph execution (Approved: True)...

[Node: Supervisor Evaluation] Polished and validated final response.
=====
FINAL SUPPORT RESPONSE:
Certainly! To initiate the refund process, please provide me with the details of your annual subscription, including the date of purchase or billing date. Once I have this information, I can assist you in processing the refund according to our refund policy. Please let me know if there are any specific steps or forms required for this process.
=====
```

5. Final Response Generation

Final support responses compiled, evaluated, and polished by the Supervisor node for delivery to the customer.

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--- Running Demo Query 5 of 5 ---
=====
CUSTOMER: "What was my previous support issue?"
=====

[Node: Classify Intent] classified query: 'What was my previous support issue?' -> Category: Memory (Rule-based)
[Node: Memory Recall] Retrieved 8 past messages from SQLite session.
=====
FINAL SUPPORT RESPONSE:
The customer's previous support issue was with their application crashing whenever they uploaded a file. The technical agent provided several troubleshooting steps, including checking the file size limit, verifying extension, clearing browser cache and trying again, updating client software, checking for corrupted files, and contacting support if necessary.
=====
```